Remaining Functionality To Be Added

BUIDCO Project Monitoring Dashboard · build backlog against RFP Clause 5 · 3 October 2026

Every item below is **not yet built**. Each traces to a clause of the BUIDCO RFP (pages 22–23, "Development of Integrated Project Monitoring Dashboard"), and each is broken down into the database, API and UI work it actually needs, based on the current schema.

8	6	5	2	21
CRITICAL	HIGH	MEDIUM	LOW	TOTAL ITEMS

Index of pending work

#	Functionality to add	RFP clause	Main work
C1	Hosting and availability remediation	5(b)(vii)	Ops
C2	Backup and disaster recovery	5(b)(vii)	Ops + docs
C3	GIS-enabled map visualisation	5(b)(ii)	DB + API + UI
C4	Insurance validity tracking and alerts	5(b)(iv)	DB + API + UI
C5	Automated push alerting (email/notification)	5(b)(v)	DB + API + UI
C6	Escalation matrix	5(b)(v)	DB + API + UI
C7	Claims and Disputes Register	5(b)(iv), 4(d)	DB + API + UI
C8	Risk Register	4(e)	DB + API + UI
H1	Payment lifecycle / IPC certification tracking	5(b)(iii), 4(c)	DB + API + UI
H2	Formal Red-Amber-Green (RAG) framework	5(b)(v)	Backend + UI
H3	Customisable report builder	5(b)(vi)	DB + API + UI
H4	Trend analysis / S-curve	5(b)(vi)	API + UI
H5	User Manual and SOPs	5(b)(viii)	Docs
H6	Monthly/quarterly Contract Management Report	4(e)	Backend + UI
M1	MIS/ERP integration capability	5(a)	API
M2	Data governance protocol	5(a)	Docs
M3	Training programme delivery	5(b)(viii)	Process
M4	Support model and SLA	5(b)(viii)	Process
M5	Security review and penetration test	5(b)(vii)	Ops
L1	Requirement assessment record	5(a)	Docs
L2	Bundle size / performance optimisation	—	Frontend

Good news first. Already built and not repeated here: the project register, sector/district/division views, milestone tracking, planned-vs-actual progress, delay analysis, Funds & UC, cost and time overrun indicators, CoS/EoT, PBG expiry alerts, O&M expiry tracking, Excel/PDF/PPT export, role-based access control and the field-level audit trail. The backlog below is what sits on top of that.

Critical / Mandatory — named in the RFP, currently absent

C1 Hosting and availability remediation

RFP 5(b)(vii) · Ops · quickest win

Gap: the API sleeps after ~15 minutes idle on its current free tier. The next request takes 33–56s (measured repeatedly) and the frontend proxy gives up first, so users see a failure. This fails "secure hosting environment" on availability grounds.

OPS	Move the API to an always-on instance, or keep it warm with an external uptime monitor pinging `/api/health` every 5 minutes.
DOCS	State an availability target and record uptime evidence.
FRONTEND	*Optional:* extend the cold-start retry already in the login screen into `baseQuery.ts`, so the whole app degrades gracefully rather than only the login page.

C2 Backup and disaster recovery

RFP 5(b)(vii) · Ops + docs

Gap: no backup schedule, retention policy, restore procedure or recovery objective exists anywhere in the project. This is a live data-loss risk, not only a compliance gap.

OPS	Automated nightly database backups with defined retention; off-site/second-region copy; backup success monitoring and alerting.
DOCS	Written restore runbook, stated RPO and RTO, and a **tested** restore (a restore that has never been run is not a backup).

C3 GIS-enabled map visualisation

RFP 5(b)(ii) · largest build

Gap: no map, no mapping library, and **no coordinate data to plot** — `geo_photo` holds only url, caption, date, source and filename. The "Geo Photos" route is a placeholder marked *deferred*.

DATABASE	Add `latitude` / `longitude` (numeric 9,6) to `project` and to `geo_photo`. Optionally district centroids as a fallback for projects without a surveyed point.
API	Return coordinates in the project list payload; add a map-data endpoint filtered by sector, division and status.
UI	Add a mapping library; new Map page in the sidebar replacing the deferred placeholder; markers coloured by status (shares the RAG scheme from H2); filter panel; marker click opens the project profile.
DATA	Back-fill coordinates for existing projects — a field/data exercise that should start early, independently of the build.

C4 Insurance validity tracking and alerts

RFP 5(b)(iv) · mirrors existing PBG work

Gap: the clause names performance security *and insurance*. PBG is handled well; insurance has no field, no alert and zero references in the codebase.

DATABASE	Add to `project`: `insurance_policy_no`, `insurer`, `insured_amount_cr`, `insurance_start_date`, `insurance_expiry_date`. Add a `v_insurance_expiry_alerts` view mirroring the existing `v_pbg_expiry_alerts`.
API	Expose the fields on the project payload; add an insurance-expiry alerts endpoint alongside the PBG one.
UI	Fields in the Contract & Security section; "Insurance Expiring ≤ 30 Days" dashboard card beside the PBG card; an Insurance Alert column mirroring the PBG alert cell.

The PBG implementation is a complete working template for this — copy its shape rather than inventing a new one.

Critical / Mandatory (continued)

C5 Automated push alerting

RFP 5(b)(v) · unlocks C6

Gap: the conditions are already detected and shown on screen — delays, cost overruns, expiring PBG and O&M, milestones. What is missing is delivery: there is **no scheduler anywhere in the backend** and nothing is ever pushed to a user.

BACKEND	Introduce a scheduled job runner (none exists today). Evaluate alert rules on a schedule and record what was sent, so an alert fires once rather than every run.
DATABASE	`notification` table: type, project, severity, message, recipient, sent_at, read_at.
DELIVERY	Email is ready — SMTP via nodemailer is already wired for password-reset OTP. **SMS is a stub only**; a gateway (e.g. Twilio) must be chosen if SMS alerts are required.
API / UI	Notifications list and mark-read endpoints; a notification bell and panel in the top navigation; per-user alert preferences.

C6 Escalation matrix

RFP 5(b)(v) · depends on C5

Gap: no escalation rules, severity tiers, owner hierarchy or time-based routing.

DATABASE	`escalation_rule`: trigger type, threshold, severity, level, target role or user, delay before escalating. `escalation_event`: an audit log of what escalated, to whom and when.
API	Rule CRUD restricted to MD/Admin; evaluation runs inside the C5 scheduler.
UI	Escalation Matrix configuration page; escalation history shown on the project profile.

C7 Claims and Disputes Register

RFP 5(b)(iv) + 4(d)

Gap: required by two clauses. Today "dispute" exists only as free text inside remark fields — there is no structure to report on.

DATABASE	`contract_claim`: project, claim no, type, claimed amount, date raised, status, decision, decided date, remarks. `contract_dispute`: project, dispute no, subject, stage (negotiation / conciliation / arbitration / litigation), value, date raised, status, next hearing date. Link claims to `cos_eot_item` where a claim led to a variation or EoT.
API	Nested CRUD under the project, following the existing CoS/EoT route pattern.
UI	Claims and Disputes sections in the Input Sheet (mirror the CoS/EoT section); a register page; open-claim and open-dispute counts on the project profile and Overview.

C8 Risk Register

RFP 4(e)

Gap: an explicit consultant deliverable with no system support at all.

DATABASE	`project_risk`: project, risk code, category, description, likelihood (1–5), impact (1–5), derived score, mitigation, owner, status, review date.
API	Nested CRUD under the project; portfolio-level top-risks query.
UI	Risk section in the Input Sheet; Risk Register page; a likelihood/impact heat map and a "top risks" card on the Overview dashboard.

High Priority

H1 Payment lifecycle / IPC certification tracking

RFP 5(b)(iii) + 4(c) · highest real-world risk

Gap: payment data is summary-only — total paid, last payment date, last RA bill number. There is no bill-by-bill record and no certification workflow, which is what Clause 4(c) describes (IPC scrutiny and final bills).

DATABASE	`interim_payment_certificate`: project, IPC no, period from/to, gross amount, deductions, net payable, submitted / verified / certified / paid dates, status, remarks.
API	Nested CRUD plus an ageing query — pending bills bucketed by days outstanding.
UI	IPC section in the Input Sheet; a Payments page with ageing buckets; pending-bill ageing feeds the C5 alert engine.

Treat this as higher risk than its priority label suggests — late contractor payments are what caused Bihar's earlier state works system to be suspended within a month of launch.

H2 Formal Red-Amber-Green (RAG) framework

RFP 5(b)(v)

Gap: colour coding exists and is used consistently, but there is no defined three-state model, no documented thresholds and no roll-up — so it is not a defensible RAG scheme.

BACKEND	Define thresholds per metric (schedule, cost, progress) — ideally configurable rather than hard-coded; compute a RAG state per project and a portfolio roll-up.
UI	Consistent R/A/G chips across table, profile and map; a visible legend; a portfolio RAG tile on the Overview; filter projects by RAG state.

H3 Customisable report builder

RFP 5(b)(vi)

Gap: "customizable" currently means ad-hoc filtering and column show/hide. Nothing can be named, saved, re-run or shared.

DATABASE	`report_definition`: name, owner, filters, selected columns, grouping, shared flag.
API / UI	Save-current-view-as-report; My Reports page; re-run and export in the existing Excel/PDF formats; optional scheduled delivery via C5.

H4 Trend analysis / S-curve

RFP 5(b)(vi) · data already exists

Gap: history is captured but never shown as a trend. The physical-history and milestone-history endpoints already exist — this is largely a visualisation task.

API	Aggregate time-series endpoint at project and portfolio level, using the existing history data and the MPR month field.
UI	Planned-vs-actual S-curve; period-on-period movement; sparklines in the projects table.

H5 User Manual and SOPs

RFP 5(b)(viii) · docs

Gap: an in-app KPI guide exists and is useful, but there is no role-wise manual and no SOP set.

DOCS	Role-wise user manual (MD / Admin / PD / Viewer); SOPs for data entry, approval and month-end close; link them from the existing in-app guide drawer.

Inadequate training is the second documented reason the earlier state system failed — this is not box-ticking.

H6 Monthly / quarterly Contract Management Report

RFP 4(e)

Gap: a named periodic deliverable with no scheduled generation.

BACKEND / UI	Standard report template; scheduled generation via the C5 scheduler; a downloadable archive of past periods.

Medium Priority

M1 MIS/ERP integration capability

RFP 5(a) · confirm scope first

The clause is qualified "if any" — establish what BUIDCO actually runs before building anything.

API Machine-to-machine authentication (API key or service account) separate from user JWTs; a published OpenAPI specification; inbound/outbound sync endpoints as required.

M2 Data governance protocol

RFP 5(a) · docs

Validation is strong (schema checks at the API edge, reused by the bulk importer); governance is undocumented.

DOCS Data dictionary, field ownership, correction policy, retention schedule, approval chain for master-data changes.

M3 Training programme delivery

RFP 5(b)(viii) · process

PROCESS Training plan and materials, role-wise sessions, attendance record. Nominate nodal officers who genuinely know departmental and field-level procedure — the previous system's nodal officers attended training but could not then train their own staff.

M4 Support model and SLA

RFP 5(b)(viii) · process

PROCESS Support channel, response and resolution targets, cover during the agreed maintenance period, escalation contacts.

M5 Security review and penetration test

RFP 5(b)(vii) · ops

Current posture is reasonable — HTTPS, security headers, CORS allowlist, bcrypt hashing, rotating refresh tokens, rate limiting. What is missing is independent evidence.

OPS Hardening checklist, dependency vulnerability audit, third-party penetration test and remediation record.

Low Priority / Enhancement

L1 Requirement assessment record

RFP 5(a) · docs

DOCS Documented requirement assessment signed off with BUIDCO. Retrospective paperwork that closes an audit point cheaply.

L2 Bundle size and performance optimisation

not an RFP requirement

FRONTEND The main JavaScript bundle is ~1.2 MB; code-split the heavy export libraries (Excel, PDF, PowerPoint) so they load on demand. Improves usability on slow government connections.

Build Order & Tracking

Sequenced by dependency and by what removes risk earliest

Recommended sequence

Stage	Items	Why in this order
1 · Stabilise	C1, C2, M5	Platform hygiene, independent of features. C1 is the cheapest item on the whole list and removes a failure a client could see in a demo. Do this before showing the system to anyone.
2 · Close data gaps	C4, C7, C8	Additive tables and fields next to structures that already exist (PBG, CoS/EoT). Low risk to current behaviour, and each one closes a named clause.
3 · Make alerts real	C5, C6, H2	Must follow stage 2 so insurance, claims and risks can also raise alerts. Detection logic already exists, so this is mostly delivery and rules.
4 · Geospatial	C3	Largest build and needs coordinate capture plus back-filling. Start collecting coordinates during earlier stages so the data is ready when the map is.
5 · Financial depth	H1, H6	Heaviest workflow change. Sequenced after alerting so pending-bill ageing can feed the alert engine on day one.
6 · Analytics	H3, H4	Best built once every entity exists, so reports and trends can span all of them.
7 · Handover	H5, M2, M3, M4, L1, L2	Documentation and enablement must describe the finished system — but draft them alongside the build, not in the final week.

Dependencies to respect

- **C6 requires C5** — escalation runs on the alert scheduler, so the scheduler must exist first.
- **H6 and scheduled reports in H3 require C5** — same scheduler.
- **C5 is more valuable after C4, C7 and C8** — otherwise it can only alert on the conditions already covered today.
- **H1 feeds C5** — bill ageing becomes an alert source once IPC records exist.
- **C3 shares the RAG colours defined in H2** — settle the scheme before colouring map markers.
- **C3 data collection has a long lead time** — begin it in parallel with stage 1, not at stage 4.

Progress checklist

Critical	High / Medium / Low
☐ **C1** Hosting and availability	☐ **H1** Payment lifecycle / IPC
☐ **C2** Backup and disaster recovery	☐ **H2** RAG framework
☐ **C3** GIS map visualisation	☐ **H3** Report builder
☐ **C4** Insurance validity and alerts	☐ **H4** Trend analysis / S-curve
☐ **C5** Automated push alerting	☐ **H5** User Manual and SOPs
☐ **C6** Escalation matrix	☐ **H6** Periodic contract report
☐ **C7** Claims and Disputes Register	☐ **M1** MIS/ERP integration ☐ **M2** Data governance
☐ **C8** Risk Register	☐ **M3** Training ☐ **M4** Support/SLA ☐ **M5** Security review
	☐ **L1** Requirements record ☐ **L2** Bundle optimisation

Scope note. Nothing in this document has been implemented. It is a plan derived read-only from the existing codebase and the RFP; no application file, schema object or configuration was modified in producing it.